

MANAGERIAL DECISION MAKING · GROUP TERM PROJECT

Predicting Retail Sales from Multiple Business Drivers

FreshMart Retail — a multiple linear regression case study with a BI dashboard

Team: [add group member names] · Course & Section: [add course/section] · Date: [presentation date]

Headline finding: location and stockouts move sales more than any single ad channel.

Business Problem & Objective

- **The decision:** FreshMart's marketing and operations teams disagree on where the next rupee of budget should go — more digital advertising, better in-stock availability, deeper discounts, or new-store features such as parking.
- **The core question:** which business factors most strongly explain a store's monthly sales revenue, and can we predict sales from store characteristics well enough to guide budget decisions?
- **Why it matters:** FreshMart operates stores across metro, urban and suburban catchments with materially different economics — an evidence-based answer prevents budget from being allocated by intuition or department politics.
- **What success looks like:** a regression model that identifies statistically significant drivers, quantifies their effect size, and supports a concrete what-if budget recommendation.

Project Objectives

- Frame the business question as a prediction problem with one dependent variable (Monthly_Sales_K) and several independent business drivers.
- Explore relationships visually and numerically before modelling — scatter plots, correlation, summary statistics.
- Prepare the data correctly: encode categories, exclude identifiers, split into train/test sets, with every assumption stated explicitly.
- Fit and interpret a multiple linear regression using scikit-learn and statsmodels OLS, including significance testing.
- Communicate findings through an interactive BI dashboard and a structured group presentation.
- Generate a prediction for a realistic what-if scenario, recommend an action, and state the model's limits honestly.

Dataset Overview

Unit of observation: one store in one month (a “store-month”). Source: simulated business data.

300

Store-months

8

Predictors used

1

Target variable

Rs '000

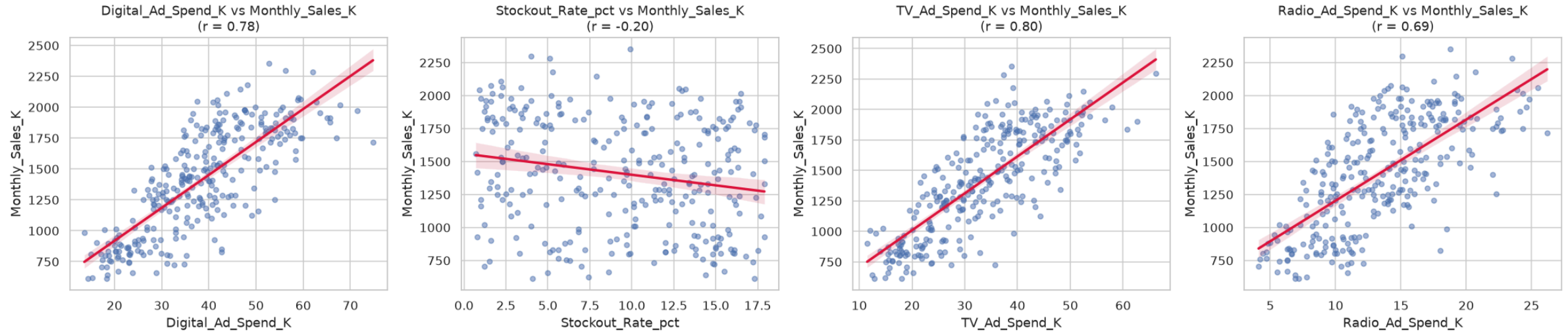
Sales unit

- **Target — Monthly_Sales_K:** monthly sales revenue per store (Rs '000).
- **Key drivers used:** Location_Type (Metro/Urban/Suburban), Is_Festival_Month, Parking_Available, TV/Digital/Radio/Print Ad Spend, Stockout_Rate_pct.
- **Assumption stated explicitly:** the data file available for this run carried 8 of the 20 predictors documented in the original project data dictionary. All 8 available drivers were used; the missing ones are treated as omitted variables in the reflection (Slide 13).

Data Cleaning & Preparation

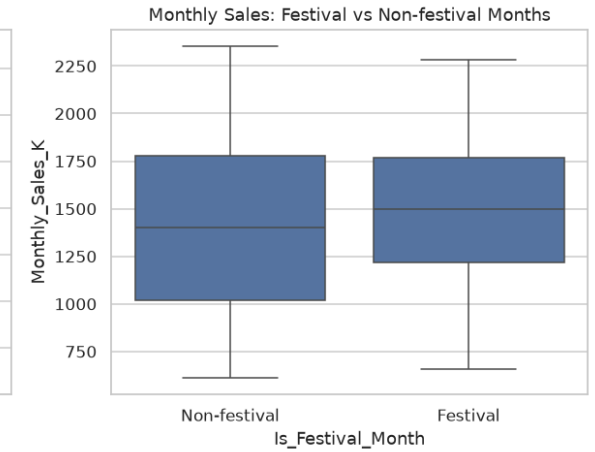
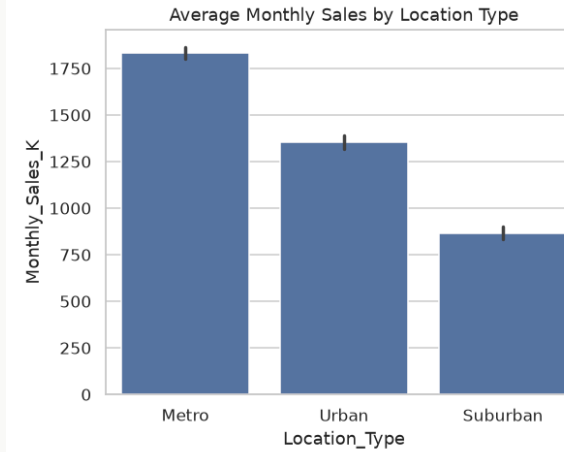
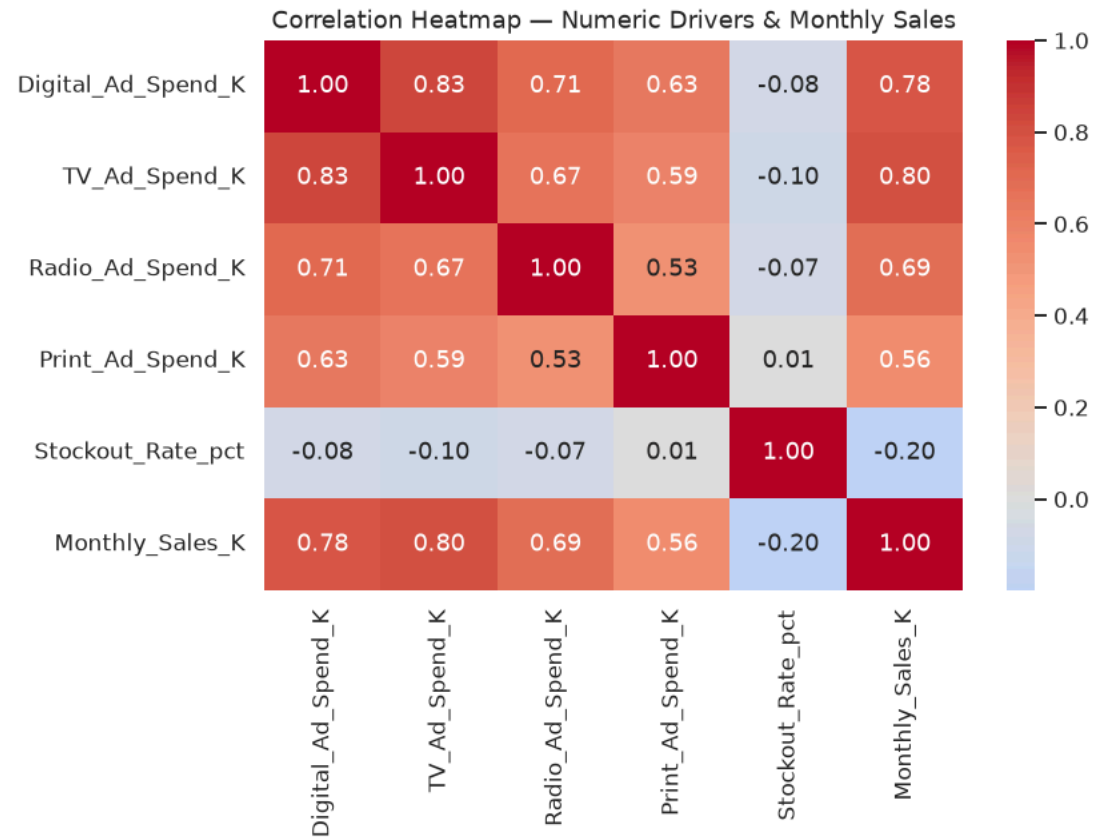
- **Missing values:** none found across all 300 rows (imputation logic included in code as a safeguard even though not needed here).
- **Duplicates:** 0 duplicate store-months (checked on all columns excluding Store_ID/Month).
- **Outlier scan:** IQR rule flagged 2 high-spend stores as statistical outliers; both retained (report-only) — residual diagnostics were used instead of ad hoc deletion.
- **Identifiers excluded:** Store_ID and Month dropped — no predictive content.
- **Categorical encoding:** Location_Type one-hot encoded, dropping one level (Metro) to avoid the dummy-variable trap.
- **Train/test split:** 75% / 25% (225 / 75 rows), random_state = 42 for full reproducibility.
- **Scaling:** not applied to the reported model — raw-unit coefficients stay interpretable; a standardized version is used only to rank driver importance (Slide 10).

Exploratory Data Analysis



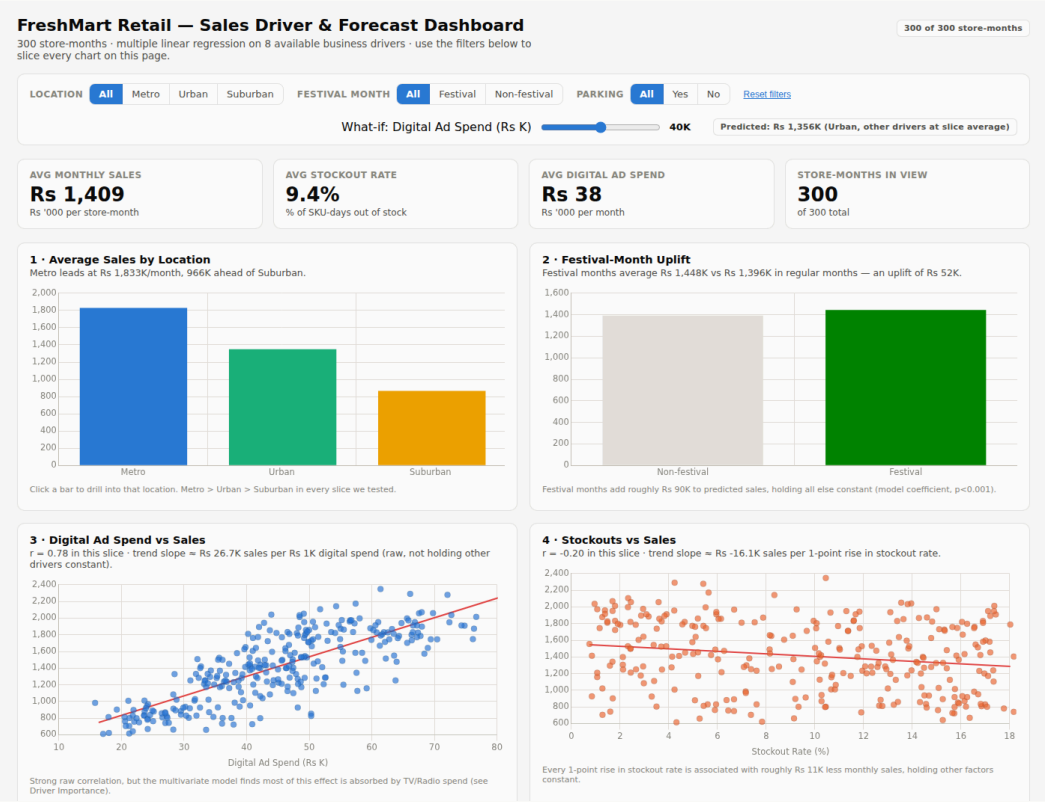
- **Digital & TV ad spend:** strong positive relationships with sales ($r = 0.78$ and $r = 0.80$).
- **Radio spend:** also strongly positive ($r = 0.69$), moving in step with TV/Digital.
- **Stockout rate:** weak negative relationship in isolation ($r = -0.20$) — its true effect is much clearer once other drivers are controlled for (Slide 10).

Important Visualizations



Metro > Urban > Suburban at every filter combination; festival months show a visibly higher median and wider upper spread.

Dashboard Overview



- 10 required items: KPI cards, location & festival comparisons, 2 driver scatter+trend charts, correlation heatmap, driver-importance tornado, predicted-vs-actual, residual histogram, ad-channel mix.
- 3 working filters (Location, Festival month, Parking) plus a live what-if Digital Ad Spend slider, with one-click drill-down from the location bar chart.

Key Insights

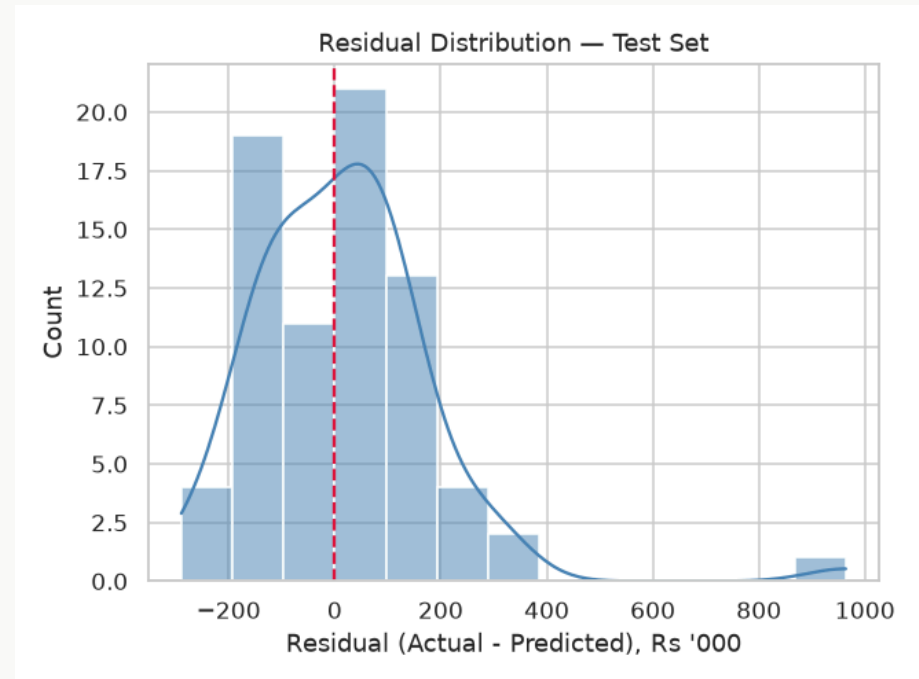
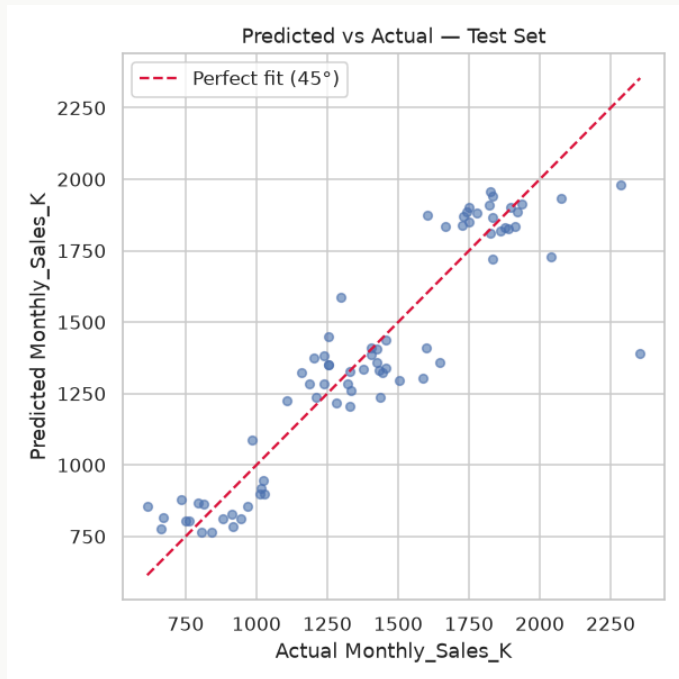
- **Location is the single biggest lever:** Metro stores out-sell Suburban stores by ~Rs 750K/month in the model.
- **TV & Radio carry the ad-budget signal, not Digital:** Digital's strong raw correlation ($r \approx 0.78$) becomes statistically insignificant ($p = 0.93$) once TV and Radio are in the model — they're highly collinear ($VIF \approx 4.5$).
- **Stockouts are a quantifiable, controllable revenue leak:** each 1-point rise in stockout rate costs ~Rs 11K/month per store ($p < 0.001$).
- **Festival months add a reliable, budget-independent uplift:** ~Rs 90K on average.

Statistical Findings — Coefficient Table

Feature	Coefficient (Rs K)	p-value	Significant (5%)
Is_Festival_Month	+90.75	0.000	Yes
Parking_Available	+35.65	0.096	No
Radio_Ad_Spend_K	+7.95	0.019	Yes
TV_Ad_Spend_K	+6.40	0.001	Yes
Digital_Ad_Spend_K	+0.16	0.929	No
Print_Ad_Spend_K	-0.43	0.891	No
Stockout_Rate_pct	-11.16	0.000	Yes
Location_Type_Urban	-383.47	0.000	Yes
Location_Type_Suburban	-753.63	0.000	Yes

- Reference level: Metro (dropped in one-hot encoding).
- Interpretation: each coefficient is the effect of a 1-unit change, holding all other drivers constant.
- Location and stockout rate are the strongest, most significant effects in the model.

Model Results & Fit

**0.827** R^2 **0.803**Adj. R^2 **Rs 173K**

RMSE

Rs 122K

MAE

Business Recommendations

- **1. Rebalance ad budget from Digital toward TV/Radio** for digital-heavy stores, pending a controlled test — Digital's apparent lift likely rides on overall marketing budget size, not its own channel effect.
- **2. Fund stockout reduction before further ad-spend increases.** Cutting stockout rate from ~15% to ~5% is worth an estimated ~Rs 112K/month per store.
- **3. Prioritize new-store investment in Metro/Urban catchments** over Suburban at parity spend — the location effect compounds with every other driver.
- **4. Plan inventory & staffing around festival months** using the ~Rs 90K predictable uplift as a baseline.

Limitations & Reflection

- **Multicollinearity:** TV and Digital ad spend move together ($VIF \approx 4.5$) — inflates standard errors and makes individual coefficients less trustworthy even though overall model fit stays valid.
- **Omitted variables:** only 8 of the 20 documented predictors were available in this run's data file — the reported R^2 should be read as a lower bound on what a full 20-predictor model would achieve.
- **Correlation vs causation:** a positive ad-spend coefficient shows association, not proof of causation — head office may allocate bigger budgets to stores it already expects to perform well.
- **What more data would help:** store-level footfall/SKU detail, a longer time series (for store fixed effects), and a randomized ad-spend design.

Future Scope

- Obtain the full 20-predictor dataset to close the omitted-variable gap.
- Run a randomized or quasi-experimental ad-spend test to separate causal lift from allocation bias.
- Extend to a panel/time-series model with store fixed effects.
- Rebuild the dashboard natively in Tableau or Power BI Desktop (field-by-field guide included in the submission).
- Explore regularized regression (Ridge/Lasso) or tree-based models as a robustness check given the multicollinearity identified.

Conclusion

- **The model works:** explains 83% of test-set variance in monthly sales ($RMSE \approx 12\%$ of average sales) — good enough to guide budget decisions.
- **The headline answer:** location and operational execution (stockout rate) explain more of the sales gap between stores than any single advertising channel does.
- **The honest caveat:** Digital ad spend's apparent effect is largely a multicollinearity artifact, not a proven independent driver.

Bottom line: fix stockouts and lean into Metro/Urban growth before chasing more digital ad spend.

Thank You

Questions & Discussion

Backup slides available on request: full coefficient table, VIF multicollinearity table, additional dashboard views, and the what-if scenario walkthrough.